Lab Course: Interaction and Correlations
Assignment #1 – Data Collection Templates

University of Bologna

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1 Overview

This standalone document contains the structured table templates required for reporting the findings of Assignment #1. The text, headers, and captions are fully formatted in English using the `booktabs` package conventions (`\toprule`, `\midrule`, and `\bottomrule`) to meet professional academic publishing standards.

2 Activity 1: Monte Carlo Estimate of π

2.1 Convergence of π Estimate with Sample Size N

This table is designed to track and compare the convergence performance of the Area Method against the direct MC Estimator as the total number of independent random samples N increases. It evaluates how closely each method approaches the analytical value of π and how the variance scales with sample size.

Table 1: Convergence tracking of the π estimate and numerical variance across different sample sizes (N).

Number of Samples (N)	Method	Mean (π Estimate)	Variance
1,000	Area Method		
	MC Estimator		
10,000	Area Method		
	MC Estimator		
50,000	Area Method		
	MC Estimator		

2.2 Statistical Comparison over M Independent Runs

By setting $N = 50,000$ and executing $M = 500$ independent trials with distinct random number sequences, we reconstruct the full distribution of estimators. This table synthesizes the global mean and the empirical variance of the sampled distributions, highlighting the structural precision advantages of direct function sampling over binary counting.

Table 2: Statistical properties and variance comparison of the sampled distributions for $N = 50,000$ and $M = 500$.

Method	Mean of the Distribution	Variance of the Distribution
Area Method		
MC Estimator		

3 Activity 2: Markov Chain Monte Carlo (MCMC)

3.1 MCMC Convergence and Acceptance Rate

This table monitors the evolution of the Markov Chain during the sampling of a bimodal Gaussian distribution under default parameters. It tracks the acceptance-to-rejection ratio across various lengths of the chain (N) alongside the Mean Squared Error (MSE) evaluated against the exact analytical distribution curve.

Table 3: MCMC update statistics and Mean Squared Error (MSE) as a function of the total runtime samples (N).

Number of Samples (N)	Accepted Updates (%)	Rejected Updates (%)	MSE (vs Analytical)
1,000			
10,000			
50,000			
100,000			

3.2 MCMC Parameter Exploration

This template is dedicated to multi-variable parameter exploration. It documents how modifying the internal structure of the target bimodal Gaussian distribution (centers x_0, standard deviations σ) and clipping its physical boundaries ($x_{\min}$, $x_{\max}$) affects the Metropolis-Hastings acceptance rate during a fixed production run.

Table 4: Metropolis acceptance rate dependencies under custom distribution profiles ($N = 50,000, N_{\text{therm}} = 5,000$).

x_0 (Center)	σ (Std Dev)	$x_{\min}$	$x_{\max}$	Acceptance Rate (%)	Notes / Configuration Typer
Default	Default	Default	Default		Baseline initialization
3.0	1.0	-10.0	10.0		Symmetrical expanded domain
2.0	1.0	-4.0	4.0		Truncated bimodal tail profile